Microbiology Galaxy Lab (MGL): The first community-driven gateway for reproducible and FAIR analysis of microbiological data analysis

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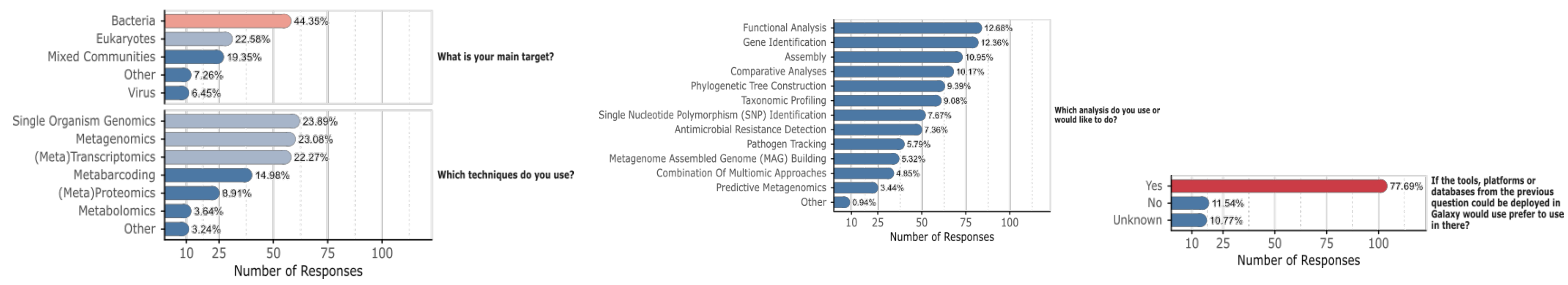
Microbial research increasingly relies on high-throughput and multi-omics approaches, generating complex datasets. However, identifying and navigating relevant tools and workflows within platforms like Galaxy remains challenging

How do scientists analyse microbial data?

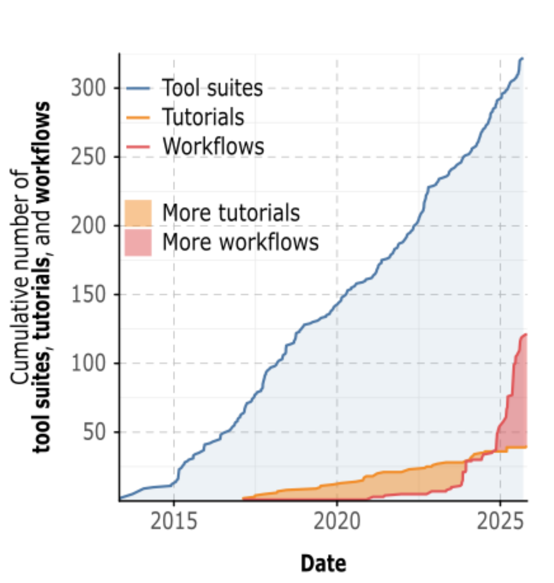
Scientists study different organisms using different techniques and analyses, and want to use Galaxy

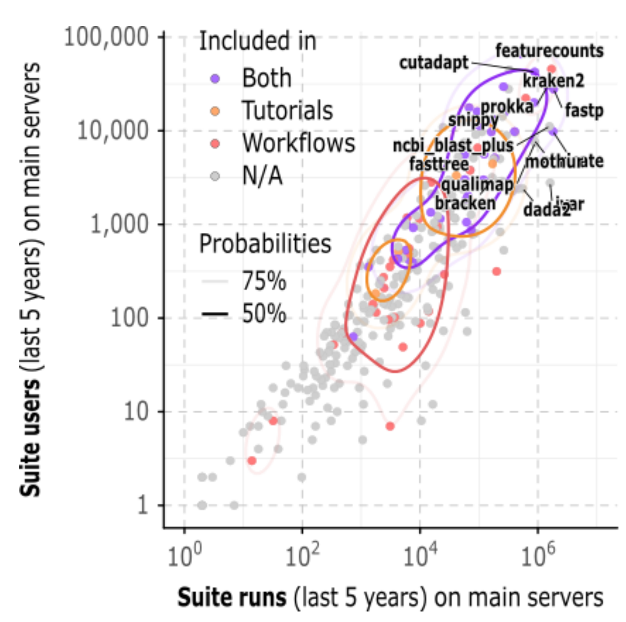


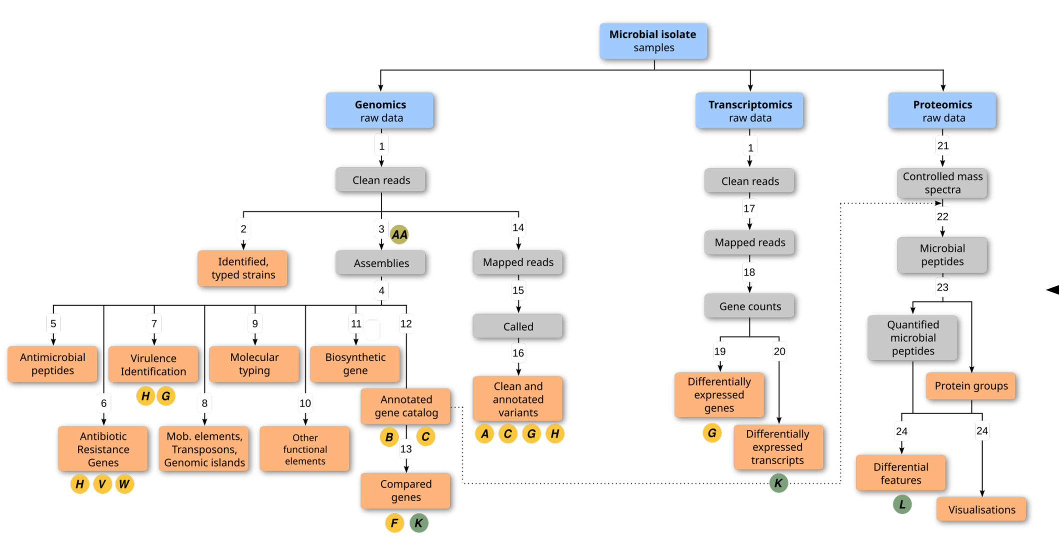
ResReplyOpenults of a survey conducted among the microbiology research community (March - September 2023, 130 responses)

Could scientists use Galaxy for microbial data analysis?

Resources for microbial analysis in Galaxy are highly used for a wide range of key computational functions

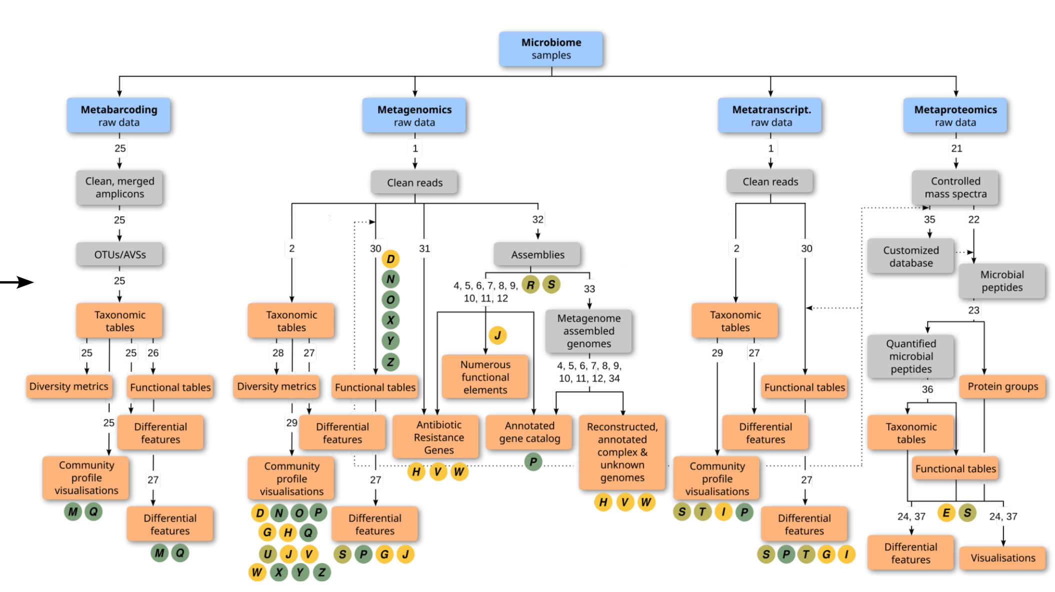






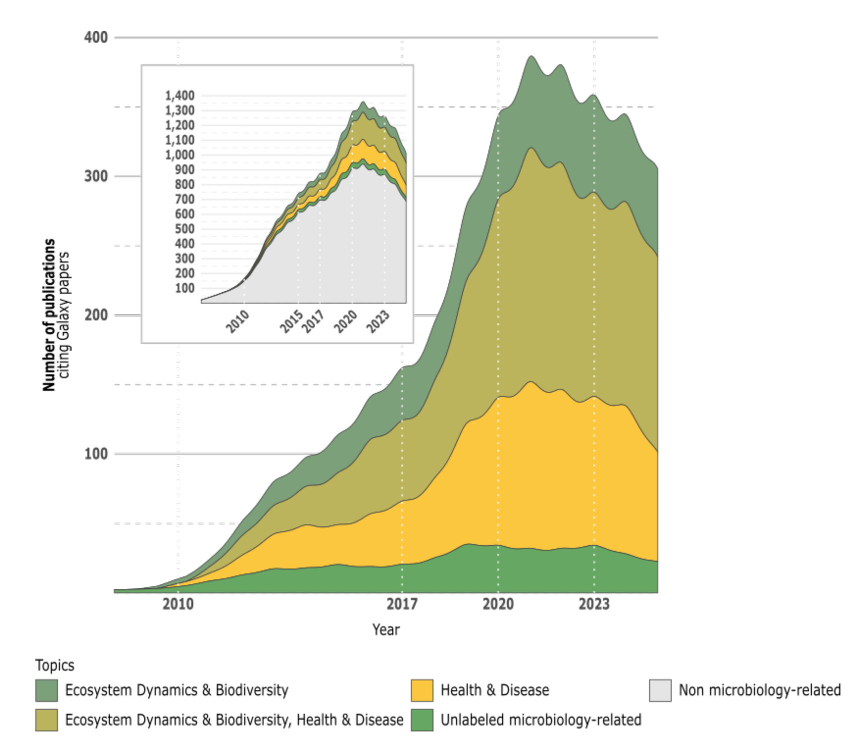
Single-organism

Microbiome

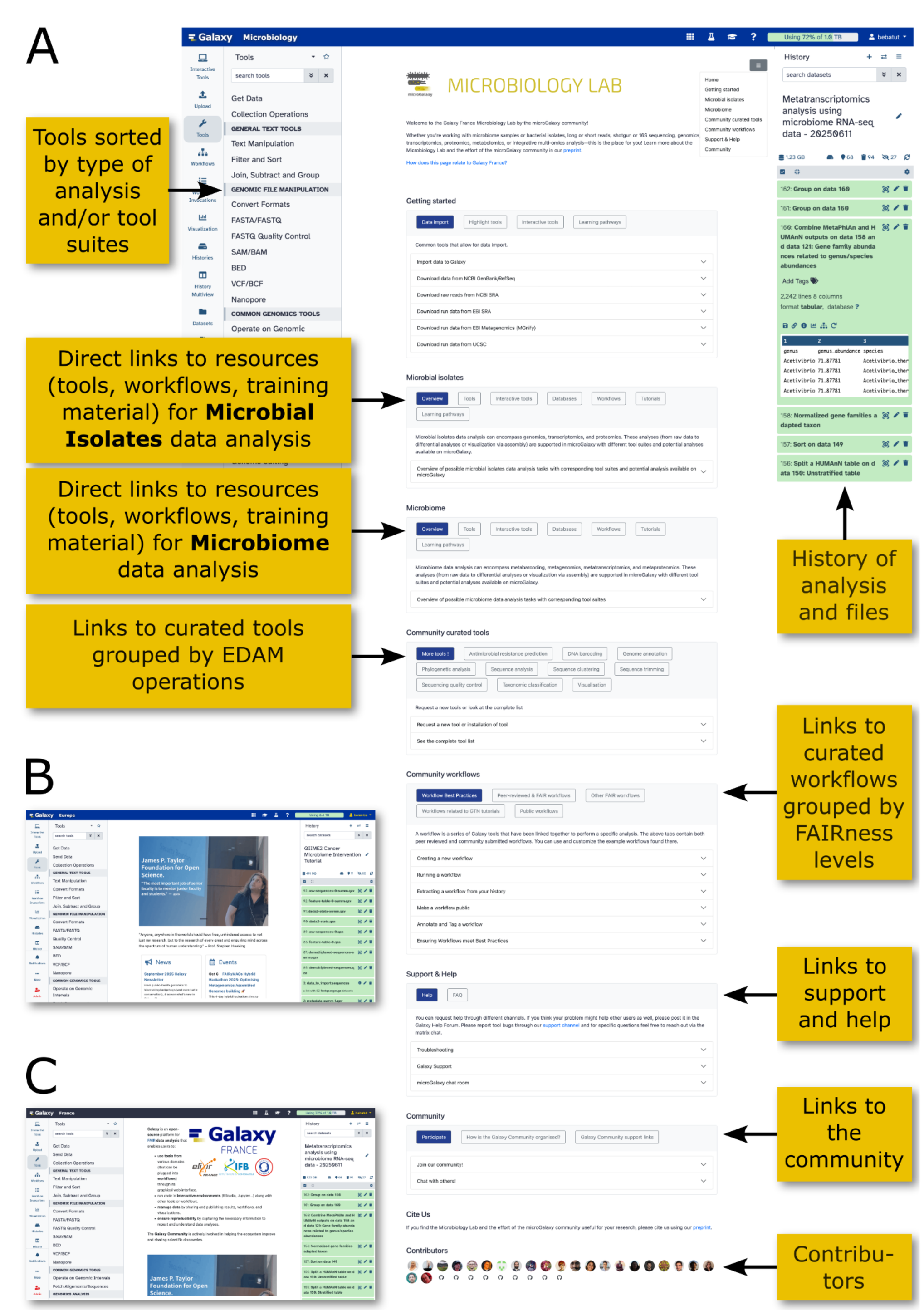


Do scientists use Galaxy for publications?

28% of publications citing Galaxy are microbiology-related papers



How does MGL help scientists find the right resources for microbial data analysis?



MGL has:

- 884 microbiology related tools
- 117 workflows
- 39 tutorials

MGL is FAIR:

- Findable (e.g., WorkflowHub, Dockstore)
- Accessible (Galaxy servers: org, .eu, .org.au, .fr)
- Interoperable (EDAM)
- Reusable (e.g., versioned workflows)

MGL is sustainable:

- microGalaxy Community-driven ecosystem
- Roadmap 2026–2030

Availability

microbiology.usegalaxy.org, .eu, .org.au ,.fr

Docker image

bioRxiv THE PREPRINT SERVER FOR BIOLOGY

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